Embryology and HnN: Development of Face & Pharyngeal Arches (Set 1)

1. The early development of the face centers around a primitive depression lined by ectoderm that forms the future oral cavity. This primitive pit is the:
a. Proctodeum
b. Pharyngeal pouch
c. Otic vesicle
d. Blastocyst cleft
e. Stomodeum (Stomatodeum)
2. At the beginning of the fourth week, the stomodeum is separated from the primitive foregut by a thin tissue barrier. This barrier is the:
a. Anal membrane
b. Cloacal membrane
c. Buccopharyngeal (oropharyngeal) membrane
d. Pleuroperitoneal membrane
e. Branchial cuticle
3. The primordial face is formed by five distinct mesenchymal prominences. Which prominence sits unpaired in the midline, forming the superior boundary of the stomodeum?
a. Maxillary prominence
b. Mandibular prominence
c. Frontonasal prominence
d. Medial nasal prominence
e. Lateral nasal prominence
4. The first pharyngeal arch splits on either side to form two essential facial prominences. These two paired structures are the:
a. Frontonasal and mandibular prominences
b. Maxillary and mandibular prominences
c. Medial and lateral nasal prominences
d. Mandibular and hyoid prominences
e. Maxillary and frontonasal prominences
5. During the fourth week, localized thickenings of surface ectoderm appear on the frontonasal prominence to initiate respiratory development. These are the:
a. Optic placodes
b. Otic placodes
c. Lens placodes
d. Olfactory (nasal) placodes
e. Branchial placodes
6. As mesenchyme proliferates around the nasal placodes, horseshoe-shaped ridges form, converting the placodes into deep pits. The inner ridges are called the:
a. Medial nasal prominences
b. Lateral nasal prominences
c. Maxillary processes

d. Globular processes
e. Frontal swellings

7. The lateral boundaries of the nasal pits are formed by separate elevations of mesenchyme, which are formally designated as the:

a. Medial nasal prominences
b. Frontonasal columns
c. Lateral nasal prominences
d. Maxillary prominences
e. Palatine processes

8. The philtrum of the upper lip and the premaxillary part of the maxilla carrying the four incisor teeth develop from the fusion of the two medial nasal processes to form the:

a. Lateral palatine process
b. Mandibular symphysis
c. Secondary palate
d. Intermaxillary segment
e. Pterygomandibular raphe

9. The main fleshy lateral parts of the upper lip, along with the majority of the maxilla and zygomatic bones, develop directly from the:

a. Mandibular prominences
b. Frontonasal process
c. Maxillary prominences
d. Lateral nasal processes
e. Hyoid arch mesenchyme

10. The lower lip, lower jaw, and the lower part of the cheeks develop entirely from the bilateral fusion of the:

a. Maxillary prominences
b. Hyoid arches
c. Frontonasal projections
d. Mandibular prominences
e. Medial nasal swellings

11. The maxillary prominence is separated from the lateral nasal prominence by a distinct groove that marks the path of the future tear-drainage system. This is the:

a. Philtrum groove
b. Nasolacrimal groove
c. Intermaxillary cleft
d. Stomodeal sulcus
e. Palatomaxillary groove

12. Cells from the ectodermal floor of the nasolacrimal groove solidifies into a cord, sinks into the mesenchyme, and canalizes to form the definitive:

a. Parotid salivary duct
b. Submandibular duct
c. Nasolacrimal duct
d. Pharyngotympanic tube
e. External acoustic meatus

13. Every pharyngeal arch is lined externally by ectoderm, internally by endoderm, and contains a core of mesenchyme supplemented heavily by migrating:

a. Hypoblast cells

b. Primordial germ cells
c. Sclerotome rows
d. Neural crest cells
e. Splanchnic mesoderm tracks

14. The cartilage model associated with the first pharyngeal arch, which serves as a scaffold for the intramembranous ossification of the mandible, is:

a. Reichert's cartilage
b. Meckel's cartilage
c. Third arch cartilage
d. Thyroid cartilage
e. Cricoid cartilage

15. Although Meckel’s cartilage largely disappears, its dorsal-most end ossifies to form two auditory ossicles of the middle ear cavity. These are the:

a. Stapes and incus
b. Malleus and stapes
c. Malleus and incus
d. Stapes only
e. Lenticular process only

16. The first pharyngeal arch is also known as the mandibular arch. All muscles derived from its mesenchyme are motorically supplied by the nerve of the first arch, which is:

a. Maxillary nerve (V2)
b. Ophthalmic nerve (V1)
c. Mandibular nerve (V3)
d. Facial nerve (CN VII)
e. Glossopharyngeal nerve (CN IX)

17. Which of the following muscles of mastication develops from the mesenchyme of the first pharyngeal arch and is supplied by the mandibular nerve?

a. Buccinator
b. Masseter (along with temporalis, medial and lateral pterygoids)
c. Orbicularis oris
d. Platysma
e. Posterior belly of digastric

18. The second pharyngeal arch is known as the hyoid arch. The specific cartilage model associated with this arch is:

a. Meckel's cartilage
b. Reichert's cartilage
c. Thyroid cartilage
d. Cricoid cartilage
e. Epiglottic cartilage

19. Reichert’s cartilage ossifies at its dorsal end to form which auditory ossicle, along with the styloid process and lesser horn of the hyoid bone?

a. Malleus
b. Incus
c. Stapes
d. Incus and malleus
e. Tympanic ring

20. All muscles derived from the second pharyngeal arch are supplied by the nerve of the second arch, which is the:
a. Mandibular nerve (V3)
b. Facial nerve (CN VII)
c. Glossopharyngeal nerve (CN IX)
d. Vagus nerve (CN X)
e. Accessory nerve (CN XI)
21. Which muscle develops from the second pharyngeal arch mesenchyme and is supplied by the facial nerve?
a. Masseter
b. Temporalis
c. Anterior belly of digastric
d. Mylohyoid
e. Muscles of facial expression (e.g., buccinator, platysma, orbicularis oris)
22. The third pharyngeal arch cartilage gives rise to the greater horn and lower body of the hyoid bone. The muscle that develops from this arch is the:
a. Tensor veli palatini
b. Stylopharyngeus
c. Cricothyroid
d. Levator veli palatini
e. Superior pharyngeal constrictor
23. The stylopharyngeus muscle is selectively supplied by the nerve of the third pharyngeal arch, which is the:
a. Facial nerve (CN VII)
b. Vagus nerve (CN X)
c. Glossopharyngeal nerve (CN IX)
d. Mandibular nerve (V3)
e. Spinal accessory nerve
24. The cartilages of the fourth and sixth pharyngeal arches fuse together to form the skeletal framework of the larynx. These arches are supplied by branches of the:
a. Glossopharyngeal nerve (CN IX)
b. Vagus nerve (CN X)
c. Facial nerve (CN VII)
d. Hypoglossal nerve (CN XII)
e. Mandibular nerve (V3)
25. The nerve of the fourth pharyngeal arch supplies the cricothyroid muscle and pharyngeal constrictors. This nerve branch is the:
a. Recurrent laryngeal nerve
b. Superior laryngeal nerve
c. Lingual nerve
d. Chorda tympani
e. Lesser petrosal nerve
26. The nerve of the sixth pharyngeal arch supplies all the intrinsic muscles of the larynx (except cricothyroid). This nerve branch is the:
a. Superior laryngeal nerve
b. Pharyngeal branch of vagus
c. Recurrent laryngeal nerve
d. Glossopharyngeal trunk

e. Accessory motor root

27. A macroscopic birth defect occurs when the maxillary prominence fails to fuse with the medial nasal prominence on one side. This results in:

a. Median cleft lip
b. Macrostomia
c. Lateral cleft lip (Hare lip)
d. Oblique facial cleft
e. Microstomia

28. An oblique facial cleft is a rare congenital anomaly where the nasolacrimal duct is exposed as an open groove on the surface. This happens due to a total failure of fusion between the:

a. Two mandibular processes
b. Maxillary prominence and lateral nasal prominence
c. Medial and lateral nasal prominences
d. Two medial nasal processes
e. First and second pharyngeal arches

29. A child is born with an abnormally large mouth opening due to a failure of proper developmental fusion between the maxillary and mandibular processes on the lateral aspect. This condition is:

a. Microstomia
b. Macrostomia
c. Mandibulofascial dysostosis
d. Median cleft lip
e. Astomia

30. Treacher Collins syndrome (mandibulofacial dysostosis) is a developmental disorder characterized by malformations of the jaw, zygomatic arch, and ears. This syndrome arises from a failure of normal migration and proliferation of neural crest cells into which specific structures?

a. Second and third pharyngeal pouches
b. Fourth and sixth pharyngeal arches
c. First pharyngeal arch derivatives
d. Frontonasal prominence exclusively
e. Stomodeal ectodermal lining

Answer Key Table

Q 1-8	Q 9-16	Q 17-24	Q 25-30
d	c	b	b
c	d	c	c
a	d	b	c
b	b	b	b
b	c	d	b
a	a	b	c
c	b	c	
d	c	b	